MH1100 Calculus I

Handout 7 - Derivatives II

Synopsis

The rules for differentiation

- Differentiation formulas:
    - Derivative of a constant function
    - The Power Rule
    - The Constant Multiple Rule
    - The Sum Rule
    - The Difference rule
    - The Product rule
    - The Quotient Rule
- Derivatives of trigonometric functions

The simplest rules

Exercise

Use the definition of derivative to find the derivatives of the following functions

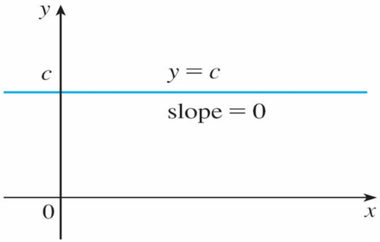
(a) $f(x) = c$ (c is a constant) (b) $f(x) = x$

a) For $f(x)=c$, where c is a constant, from the definition of derivative we have
$f'(x)= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{c-c}{h} = \lim_{h\to 0} 0 = 0$

b) For $f(x)=x$, from the definition of deviration we have
$f'(x)= \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{(x+h)-x}{h} = \lim_{h\to 0} 1 = 1$

The simplest rules

Graphically, the function $f(x) = c$ is the horizontal line $y = c$, which has slope 0, so we must have $f'(x) = 0$.



In Leibniz notation, we have the rule for a constant function as follows.

Derivative of a Constant Function

$$\frac{d}{dx}(c) = 0$$

The power rule

The function $f(x) = x$ is a special case of the power function $f(x) = x^n$ (here n is a positive integer).

For the power function $f(x) = x^n$, we have already investigated the cases $n = 1$, $n = 2$, and $n = 3$. We found that

$$\frac{d}{dx}(x) = 1, \qquad \frac{d}{dx}\left(x^2\right) = 2x, \qquad \frac{d}{dx}\left(x^3\right) = 3x^2.$$

It seems to be a reasonable guess that, when n is a positive integer,

$$\frac{d}{dx}\left(x^n\right) = nx^{n-1}.$$

$(x^3)' = \lim_{h \to 0} \frac{(x+h)^3 - x^3}{h}$
$= 3x^2$

This turns out to be true.

The Power Rule

If n is a positive integer, then

$$\frac{d}{dx}\left(x^n\right) = nx^{n-1}$$

The power rule

Before proving the Power Rule, we review the binomial theorem.

Binomial Theorem

Let a and b be two real numbers. n is a positive integer. Then any power of $a+b$ can be expanded into a sum of the form (binomial identity)

$$(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k},$$

where each $\binom{n}{k}$ is a specific positive integer known as a binomial coefficient. The binomial coefficient can be evaluated as

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \qquad \text{for } 0 \leq k \leq n$$

where $n!$ denotes the factorial of n (The product of all integers from 1 to n).

Proof of the power rule

Proof

Consider the function $f(x) = x^n$, where $n \geq 2$ is a positive integer,

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \to 0} \frac{(x+h)^n - x^n}{h}$$

Using the binomial identity,

$$
\begin{aligned}
f'(x) &= \lim_{h \to 0} \frac{(x+h)^n - x^n}{h} = \lim_{h \to 0} \frac{\overbrace{\sum_{k=0}^{n} \binom{n}{k} x^k h^{n-k}}^{n+1 \text{ terms}} - x^n}{h} \\
&= \lim_{h \to 0} \frac{\left[\underbrace{\sum_{k=0}^{n-2} \binom{n}{k} x^k h^{n-k}}_{\text{the first } n-1 \text{ terms}} + \underbrace{\binom{n}{n-1} x^{n-1} h^{n-(n-1)}}_{k=n-1} + \underbrace{\binom{n}{n} x^n \overbrace{h^{n-n}}^{h^0=1\ (h\neq 0)}}_{k=n} \right] - x^n}{h} \\
&= \lim_{h \to 0} \frac{\left[\sum_{k=0}^{n-2} \binom{n}{k} x^k h^{n-k} + \underbrace{\binom{n}{n-1}}_{n} x^{n-1} h + x^n \right] - x^n}{h} \\
&= \lim_{h \to 0} \frac{\left[\underbrace{\sum_{k=0}^{n-2} \binom{n}{k} x^k h^{n-k}}_{n-k \geq 2} + n x^{n-1} h \right]}{h} = \lim_{h \to 0} \left[\sum_{k=0}^{n-2} \binom{n}{k} x^k \underbrace{\overbrace{h^{n-1-k}}^{=0 \text{ as } h \to 0}}_{n-1-k \geq 1} + n x^{n-1} \right] \\
&= n x^{n-1}.
\end{aligned}
$$

$\hookrightarrow \therefore$ largest possible k is $n-2$

Exercise

Find the derivatives of the following functions:

(a) $f(x) = x^6$, (b) $y = x^{1000}$, (c) $y = t^4$, (d) $\frac{d}{dr}(r^3)$.

$f'(x) = 6x^5$

$y' = 1000x^{999}$

$\frac{dy}{dt} = 4t^3$

$\frac{d}{dr}(r^3) = 3r^2$

The constant multiple rule

Similar to the limits, when new functions are formed from old functions by addition, subtraction, or multiplication by a constant, their derivatives can be calculated in terms of derivatives of the old functions.

In particular, the following formula says that *the derivative of a constant times a function is the constant times the derivative of the function.*

Constant Multiple Rule

If c is a constant and f is a differentiable function, then

$\lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$ exists

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}f(x)$$

Proof. Let $g(x) = cf(x)$. Then

$$\begin{aligned} g'(x) &= \lim_{h\to 0}\frac{g(x+h)-g(x)}{h} = \lim_{h\to 0}\frac{cf(x+h)-cf(x)}{h} \\ &= \lim_{h\to 0} c\left[\frac{f(x+h)-f(x)}{h}\right] = c\lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\right] = cf'(x). \end{aligned}$$

The sum rule

The next rule tells us that *the derivative of a sum of functions is the sum of the derivatives.*

Sum Rule

If f and g are both differentiable, then

$$\frac{d}{dx}[f(x)+g(x)] = \frac{d}{dx}f(x) + \frac{d}{dx}g(x)$$

Proof. Let $F(x) = f(x) + g(x)$. Then

$$\begin{aligned}
F'(x) &= \lim_{h\to 0}\frac{F(x+h)-F(x)}{h} = \lim_{h\to 0}\frac{[f(x+h)+g(x+h)]-[f(x)+g(x)]}{h} \\
&= \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h} + \frac{g(x+h)-g(x)}{h}\right] \\
&= \lim_{h\to 0}\left[\frac{f(x+h)-f(x)}{h}\right] + \lim_{h\to 0}\left[\frac{g(x+h)-g(x)}{h}\right] \\
&= f'(x) + g'(x).
\end{aligned}$$

The sum rule and the difference rule

Using prime notation, we can write the Sum Rule as

$$(f+g)' = f' + g'$$

The Sum Rule can be extended to the sum of any number of functions. For instance, using this rule twice, we get

$$(f+g+h)' = [(f+g)+h]' = (f+g)' + h' = f' + g' + h'$$

By writing $f - g$ as $f + (-1)g$ and applying the Sum Rule and the Constant Multiple Rule, we get the following formula.

Difference Rule

If f and g are both differentiable, then

$$\frac{d}{dx}[f(x) - g(x)] = \frac{d}{dx}f(x) - \frac{d}{dx}g(x)$$

The product rule

Let's move to the following important fact

The Product Rule

If f and g are both differentiable, then

$$\frac{d}{dx}\left[f(x)g(x)\right] = f(x)\frac{d}{dx}g(x) + g(x)\frac{d}{dx}f(x)$$

Proof. Let $F(x) = f(x)g(x)$. Then

$$F'(x) = \lim_{h\to 0}\frac{F(x+h) - F(x)}{h} = \lim_{h\to 0}\frac{f(x+h)g(x+h) - f(x)g(x)}{h}$$

In order to evaluate this limit, we would like to separate the functions f and g as in the proof of the Sum Rule. This separation can be achieved by subtracting and adding the term $f(x+h)g(x)$ in the numerator:

Proof

Proof (Cont'd).

$$F'(x) = \lim_{h\to 0} \frac{f(x+h)g(x+h) - f(x+h)g(x) + f(x+h)g(x) - f(x)g(x)}{h}$$
$$= \lim_{h\to 0} \left[f(x+h)\frac{g(x+h)-g(x)}{h} + g(x)\frac{f(x+h)-f(x)}{h} \right]$$

funtion of x
independent of h

.

to use limit laws

According to the limit laws, we can split this limit into 4 pieces as long as the pieces themselves have limits as $h \to 0$. And we can check that

- $\lim_{h\to 0} f(x+h) = f(x)$ [$f(x)$ is differentiable, so $f(x)$ is continuous.]
- $\lim_{h\to 0} \frac{g(x+h)-g(x)}{h} = g'(x)$ [$g(x)$ is differentiable.]
- $\lim_{h\to 0} g(x) = g(x)$ [$g(x)$ is a constant with respect to the variable h.]
- $\lim_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$ [$f(x)$ is differentiable.]

Proof

Proof (Cont'd). Therefore, using the limit laws,

$$\begin{aligned} F'(x) &= \lim_{h\to 0}\left[f(x+h)\frac{g(x+h)-g(x)}{h} + g(x)\frac{f(x+h)-f(x)}{h}\right] \\ &= \lim_{h\to 0} f(x+h)\cdot \lim_{h\to 0}\frac{g(x+h)-g(x)}{h} + \lim_{h\to 0} g(x)\cdot \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} \\ &= f(x)g'(x) + g(x)f'(x). \end{aligned}$$

Thus,

$$(fg)' = fg' + gf'$$

In words, the Product Rule says that *the derivative of a product of two functions is the first function times the derivative of the second function plus the second function times the derivative of the first function.*

Exercise

Use the Product Rule to find $F'(x)$ if $F(x) = (2x^4)(3x^5)$.

By the Product Rule, we have

$$F'(x) = (2x^4)\frac{d}{dx}(3x^5) + 3x^5\frac{d}{dx}(2x^4)$$
$$= (2x^4)(15x^4) + (3x^5)(8x^3)$$
$$= 30x^8 + 24x^8$$
$$= 54x^8$$

Notice that we could verify the answer to this exercise directly by first multiplying the factors:

$$F(x) = (2x^4)(3x^5) = 6x^9 \Rightarrow F'(x) = 6(9x^8) = 54x^8$$

But later we will meet functions, such as $y = x\sin x$, for which the Product Rule is the only possible method.

The Quotient Rule

If f and g are both differentiable, then

$$\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)\frac{d}{dx}\left[f(x)\right] - f(x)\frac{d}{dx}\left[g(x)\right]}{[g(x)^2]}$$

Proof. Let $F(x) = f(x)/g(x)$. Then

$$\begin{aligned} F'(x) &= \lim_{h\to 0}\frac{F(x+h)-F(x)}{h} = \lim_{h\to 0}\frac{\frac{f(x+h)}{g(x+h)} - \frac{f(x)}{g(x)}}{h} \\ &= \lim_{h\to 0}\frac{f(x+h)g(x) - f(x)g(x+h)}{hg(x+h)g(x)} \end{aligned}$$

In order to evaluate this limit, we would like to separate the functions f and g as in the proof of the Product Rule. This separation can be achieved by subtracting and adding the term $f(x)g(x)$ in the numerator:

Proof

Proof (Cont'd).

$$F'(x) = \lim_{h\to 0} \frac{f(x+h)g(x) - f(x)g(x) + f(x)g(x) - f(x)g(x+h)}{hg(x+h)g(x)}$$
$$= \lim_{h\to 0} \frac{g(x)\frac{f(x+h)-f(x)}{h} - f(x)\frac{g(x+h)-g(x)}{h}}{g(x+h)g(x)}.$$

To evaluate the limit, we can check the existence of the limits

- $\lim\limits_{h\to 0} g(x) = g(x)$ [$g(x)$ is a constant with respect to the variable h.]
- $\lim\limits_{h\to 0} \frac{f(x+h)-f(x)}{h} = f'(x)$ [$f(x)$ is differentiable.]
- $\lim\limits_{h\to 0} f(x) = f(x)$ [$f(x)$ is a constant with respect to the variable h.]
- $\lim\limits_{h\to 0} \frac{g(x+h)-g(x)}{h} = g'(x)$ [$g(x)$ is differentiable.]
- $\lim\limits_{h\to 0} g(x+h) = g(x)$ [$g(x)$ is differentiable, so $g(x)$ is continuous.]

Proof

Proof (Cont'd). Therefore, using the limit laws,

$$F'(x) = \lim_{h\to 0} \frac{g(x)\frac{f(x+h)-f(x)}{h} - f(x)\frac{g(x+h)-g(x)}{h}}{g(x+h)g(x)}.$$

$$= \frac{\lim\limits_{h\to 0} g(x) \cdot \lim\limits_{h\to 0} \frac{f(x+h)-f(x)}{h} - \lim\limits_{h\to 0} f(x) \cdot \lim\limits_{h\to 0} \frac{g(x+h)-g(x)}{h}}{\lim\limits_{h\to 0} g(x+h) \cdot \lim\limits_{h\to 0} g(x)}.$$

$$= \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)^2]}.$$

Thus,

$$\left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}.$$

In words, the Quotient Rule says that *the derivative of a quotient is the denominator times the derivative of the numerator minus the numerator times the derivative of the denominator, all divided by the square of the denominator.*

Quotient Rule

Example A

Use the Quotient Rule to find y' if $y = \frac{x^2-2x+4}{x^3+8}$.

By the Quotient Rule, we have

$$y' = \frac{(x^3+8)\frac{d}{dx}(x^2-2x+4) - (x^2-2x+4)\frac{d}{dx}(x^3+8)}{(x^3+8)^2}$$
$$= \frac{(x^3+8)(2x-2) - (x^2-2x+4)(3x^2)}{(x^3+8)^2}$$
$$= \frac{(2x^4-2x^3+16x-16) - (3x^4-6x^3+12x^2)}{(x^3+8)^2}$$
$$= \frac{-x^4+4x^3-12x^2+16x-16}{(x^3+8)^2}$$

$$\frac{d}{dx}\left(\frac{1}{x+2}\right) = \frac{(x+2)(0)-1(1)}{(x+2)^2}$$

they are the same $= -\frac{1}{(x+2)^2}$

Note: Don't use the Quotient Rule every time you see a quotient. Sometimes it is easier to first rewrite a quotient to put it in a form that is simpler for the purpose of differentiation. Actually, in this example, y can be simplified into $y = 1/(x+2)$ if you can notice that $x^3+8 = (x+2)(x^2-2x+4)$.

Quotient Rule

The Quotient Rule can be used to extend the Power Rule to the case where the exponent is a negative integer.

If n is a positive integer, then

$$\frac{d}{dx}(x^{-n}) = (-n)x^{-n-1}$$

Proof

$$\begin{aligned}\frac{d}{dx}(x^{-n}) &= \frac{d}{dx}\left(\frac{1}{x^n}\right) = \frac{x^n \frac{d}{dx}(1) - 1 \cdot \frac{d}{dx}x^n}{(x^n)^2} \\ &= \frac{0 - nx^{n-1}}{x^{2n}} = \frac{-n}{x^{n+1}} \\ &= -nx^{-n-1}\end{aligned}$$

$x^{n-1} \times x^{-2n} = x^{-n-1}$
$= \frac{1}{x^{n+1}}$

The General Power Rule

Actually, the Power Rule is true for every real number.

The General Power Rule

If s is a real number, then

$$\frac{d}{dx}(x^s) = sx^{s-1}$$

The formal proof of this theorem will probably be given in Chapter 6.

Example B.

In Handout 6, by using the definition of derivative we found that

$$\frac{d}{dx}\left(\sqrt{x}\right) = \frac{1}{2\sqrt{x}}.$$

$\frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$

We can also use the Power Rule to find the derivative as follows.

$$\frac{d}{dx}\left(\sqrt{x}\right) = \frac{d}{dx}(x)^{\frac{1}{2}} = \frac{1}{2}(x)^{\frac{1}{2}-1} = \frac{1}{2}(x)^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$$

Differential Formulas

The differentiation rules enable us to find tangent lines without having to resort to the definition of a derivative.

It also enables us to find *normal lines*. The normal line to a curve C at a point P is the line through P that is perpendicular to the tangent line at P.

Example 12

Find equations of the tangent line and normal line to the curve $y = \frac{\sqrt{x}}{(1+x^2)}$ at the point $(1, \frac{1}{2})$.

Solution. According to the Quotient Rule, we have

$$\frac{dy}{dx} = \frac{(1+x^2)\frac{d}{dx}\left(\sqrt{x}\right) - \sqrt{x}\frac{d}{dx}(1+x^2)}{(1+x^2)^2}$$

$$= \frac{(1+x^2)\frac{1}{2\sqrt{x}} - \sqrt{x}(2x)}{(1+x^2)^2}$$

$$= \frac{(1+x^2) - 4x^2}{2\sqrt{x}(1+x^2)^2} = \frac{1-3x^2}{2\sqrt{x}(1+x^2)^2}$$

slope: $\left.\frac{dy}{dx}\right|_{x=1}$

Example 12–Solution (Cont'd)

Solution (Cont'd). So the slope of the tangent line at $(1, \frac{1}{2})$ is

$$\frac{dy}{dx}\Big|_{x=1} = \frac{1-3x^2}{2\sqrt{x}(1+x^2)^2}\Big|_{x=1} = \frac{1-3\cdot 1^2}{2\sqrt{1}(1+1^2)^2} = -\frac{1}{4}.$$

We use the point-slope form to write an equation of the tangent line at $(1, \frac{1}{2})$:

$y-y_0=k(x-x_0)$ $\quad k=\frac{dy}{dx}\Big|_{x=x_0}$

$$y - \frac{1}{2} = -\frac{1}{4}(x-1) \qquad \text{or} \qquad y = -\frac{1}{4}x + \frac{3}{4}.$$

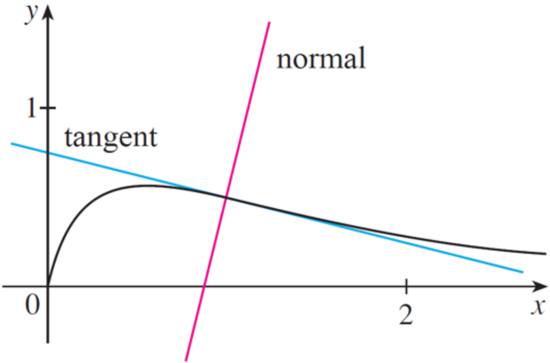
The slope of the normal line at $(1, \frac{1}{2})$ is the negative reciprocal of $-\frac{1}{4}$, namely 4, so an equation is

$y-y_0=-\frac{1}{k}(x-x_0)$

$$y - \frac{1}{2} = 4(x-1) \qquad \text{or} \qquad y = 4x - \frac{7}{2}.$$

Example 12–Solution (Cont'd)

The curve and its tangent and normal lines are graphed in the figure below.



Differentiation Formulas

We summarize the differentiation formulas we have learned so far as follows.

Table of Differentiation Formulas

$$\frac{d}{dx}(c) = 0 \qquad \frac{d}{dx}(x^n) = nx^{n-1}$$

$$(cf)' = cf' \qquad (f+g)' = f' + g' \qquad (f-g)' = f' - g'$$

$$(fg)' = fg' + gf' \qquad \left(\frac{f}{g}\right)' = \frac{gf' - fg'}{g^2}$$

Comments: Please avoid the following common mistakes.

$$(fg)' \neq f'g' \qquad \left(\frac{f}{g}\right)' \neq \frac{f'}{g'} \qquad \left(\frac{f}{g}\right)' \neq \frac{fg' - gf'}{g^2}$$

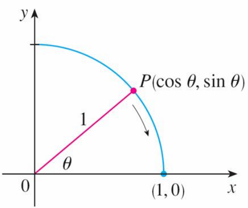
Derivatives of Trigonometric Functions

It is important to remember that when we talk about the function f defined for all real numbers x by $f(x) = \sin x$, it is understood that $\sin x$ means the sine of the angle whose radian measure is x. A similar convention holds for the other trigonometric functions.

Let's first review the following two basic facts.

We know from the definitions of $\sin\theta$ and $\cos\theta$ that the coordinates of the point P in the figure below are $(\cos\theta, \sin\theta)$. As $\theta \to 0$, we see that P approaches the point $(1, 0)$ and thus

$$\lim_{\theta\to 0} \cos\theta = 1 \qquad\qquad \lim_{\theta\to 0} \sin\theta = 0$$



Derivatives of Trigonometric Functions

Example C

Use the definition of derivative to prove that if $f(x) = \sin x$ then $f'(x) = \cos x$.

Solution. From the definition of a derivative, we have

$$
\begin{aligned}
f'(x) &= \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{\sin(x+h) - \sin x}{h} \\
&= \lim_{h\to 0} \frac{(\sin x \cos h + \cos x \sin h) - \sin x}{h} \\
&= \lim_{h\to 0} \left[\frac{\sin x \cos h - \sin x}{h} + \frac{\cos x \sin h}{h} \right] \\
&= \lim_{h\to 0} \left[\sin x \left(\frac{\cos h - 1}{h} \right) + \cos x \left(\frac{\sin h}{h} \right) \right] \\
&= \lim_{h\to 0} \underbrace{\sin x}_{} \cdot \lim_{h\to 0} \frac{\cos h - 1}{h} + \lim_{h\to 0} \underbrace{\cos x}_{} \cdot \underbrace{\lim_{h\to 0} \frac{\sin h}{h}}_{} \\
&= \sin x \cdot 0 + \cos x \cdot 1 = \cos x.
\end{aligned}
$$

constant w.r.t h

constant w.r.t h

can directly use the result

very common and well-known limit

Exercise

Prove that if $f(x) = \cos x$ then $f'(x) = -\sin x$.

$*$: $\cos(x+h) = \cos x \cos h - \sin x \sin h$

From the definition of a derivative, we have

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{\cos(x+h) - \cos x}{h}$$

$$= \lim_{h\to 0} \frac{\cos x \cos h - \sin x \sin h - \cos x}{h}$$

$$= \lim_{h\to 0} \left[\frac{\cos x \cos h - \cos x}{h} - \frac{\sin x \sin h}{h}\right]$$

$$= \lim_{h\to 0} \left[\cos x \left(\frac{\cos h - 1}{h}\right) - \sin x \left(\frac{\sin h}{h}\right)\right]$$

$$= \lim_{h\to 0} \cos x \cdot \lim_{h\to 0} \frac{\cos h - 1}{h} - \lim_{h\to 0} \sin x \cdot \lim_{h\to 0} \frac{\sin h}{h}$$

$$= \cos x \cdot 0 - \sin x \cdot 1$$

$$= -\sin x$$

Derivatives of Trigonometric Functions

We have the following fundamental formulas:

Theorem

The functions $\sin x$ and $\cos x$ are differentiable at every point of $\mathbb{R}$ and their derivatives at those points are given by the rules:

- $$\frac{d}{dx}(\sin x) = \cos x$$
- $$\frac{d}{dx}(\cos x) = -\sin x$$

Derivatives of Trigonometric Functions

The tangent function $\tan x$ can also be differentiated by using the definition of a derivative, but it is easier to use the Quotient Rule together with the formulas in the previous theorem:

$$
\begin{aligned}
\frac{d}{dx}(\tan x) &= \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) \\
&= \frac{\cos x \cdot \frac{d}{dx}(\sin x) - \sin x \cdot \frac{d}{dx}(\cos x)}{\cos^2 x} \\
&= \frac{\cos x \cdot \cos x - \sin x \cdot (-\sin x)}{\cos^2 x} \\
&= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} \\
&= \frac{1}{\cos^2 x} = \sec^2 x
\end{aligned}
$$

$\frac{1}{\cos x} = \sec x$

cosec

The derivatives of the remaining trigonometric functions, csc, sec, and cot, can also be found easily using the Quotient Rule.

Derivatives of Trigonometric Functions

We collect all the differentiation formulas for trigonometric functions in the following table. Remember that they are valid only when x is measured in radians.

*can use without proving.

Derivatives of Trigonometric Functions

$$\frac{d}{dx}(\sin x) = \cos x \qquad \frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cos x) = -\sin x \qquad \frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x \qquad \frac{d}{dx}(\cot x) = -\csc^2 x$$